The desk: the web layer

Above: DESK.md §4 (the web layer).

Talks to:

- the front door, which brings it every page and API call;
- home, the running tools and the stopped tools, whose pages and routes it serves;
- the scheduler, which runs the timed work the pages trigger.

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1. What it is

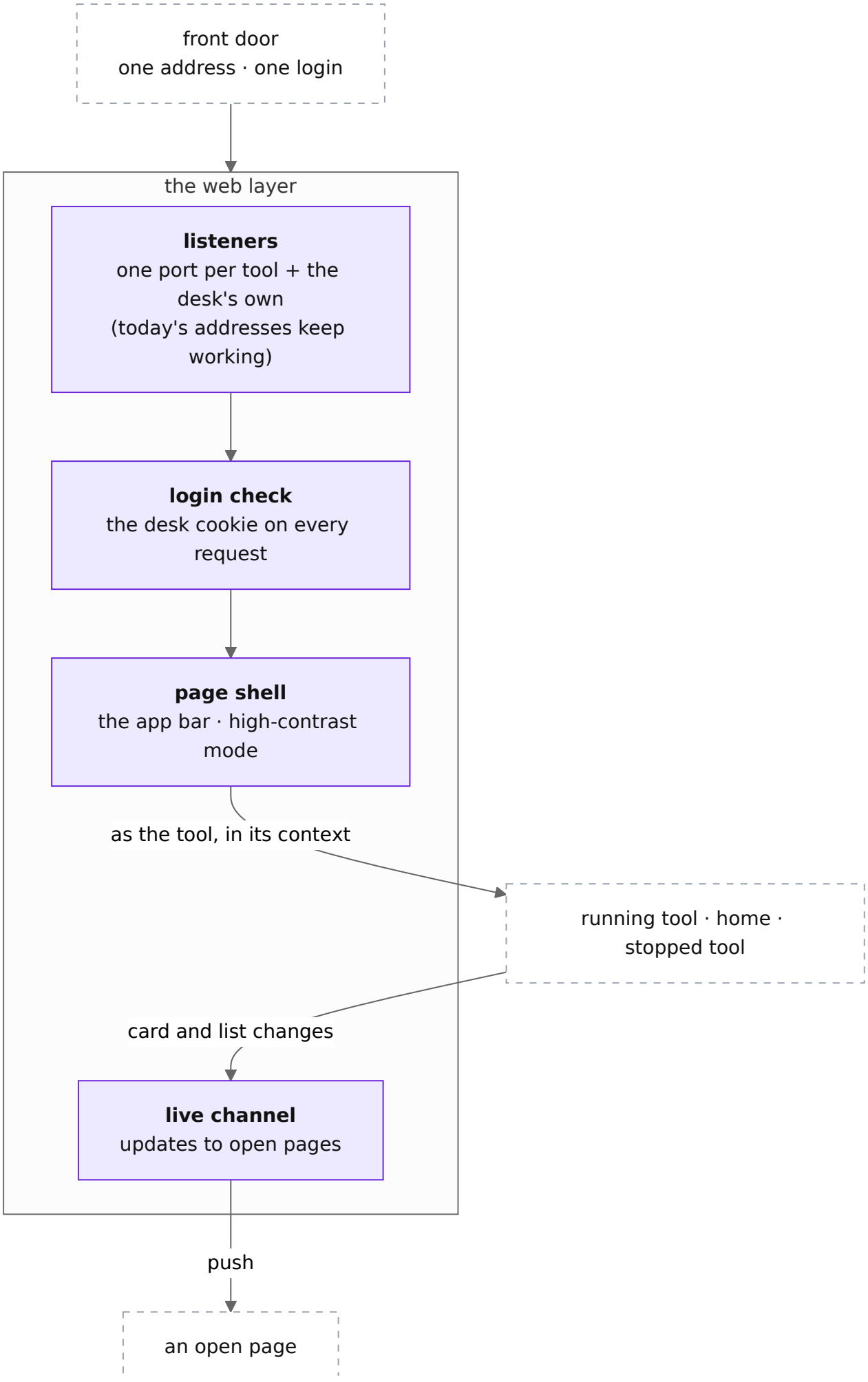
The one way in, for every tool and for home. The front door hands the desk every request; the web layer decides which tool (or home) it belongs to, checks the login, wraps it in the page shell, and **runs it inside that tool's context** so it can only touch that tool's file (DESK.md §3). It keeps today's addresses working — one listener per tool port, plus the desk's own — and pushes live updates to pages left open.

What it owns: the listeners, the login check, the page shell, and the live channel.

What it never does: scan, compute a card, or let a page run heavy work in the request (that goes to a job, DESK.md §4.2).

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2. The diagram


front door
one address · one login
the web layer
listeners
one port per tool + the desk's own (today's addresses keep working)
login check
the desk cookie on every request
page shell
the app bar · high-contrast mode
as the tool, in its context
running tool · home · stopped tool
card and list changes
live channel
updates to open pages
push
an open page

3. Its parts

3.1 One listener per tool, plus the desk's own

Today each tool is its own program on its own port, and the trader (and qp) reach it there. The desk keeps **every one of those ports**: the one program listens on each tool's port and on the desk's own, so a saved address, a bookmark, and qp's register reads over the old URLs all keep working while the tools move in one at a time (MIGRATION.md). A request on a tool's port is that tool's; a request on the desk's own port is home's.

3.2 The login check

Every request carries the desk's login cookie, checked here before anything runs — the same login for every tool, home, qp and the journal (FRONT-DOOR.md). A call from the box itself (qp reading a register, the scheduler) is allowed without it. Nothing on a tool's port is open to the outside without the cookie (S-1, closed by the login).

3.3 The page shell

Every page carries the same top bar naming the programs (screeners, qp, journal) with the current one marked, and a high-contrast mode readable outdoors. The shell is served from one place, so a change to the bar reaches every page at once.

3.4 Running each request as its tool

The web layer resolves the tool from the port and runs the request **inside that tool's context** (DESK.md §3): the page reads that tool's live list and writes only that tool's file. A database query made outside every context is an error, not a quiet write into another tool's file — so one tool's page can never reach another's results.

3.5 The live channel

A page left open is kept fresh: when a scan, a star, or a re-quote changes a tool's cards, the change is pushed to that tool's open pages, rather than the page showing numbers as old as the last load (D-S15). The channel is per tool, so a page only ever sees its own tool's updates. qp's chart has its own live tail (FEEDS.md); this is the desk's.

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4. Contracts

Served by the desk (one Node program), behind the front door. The routes themselves belong to the pages they serve:

Routes	Document
a running tool's pages and API (the card list, the market read, the warehouse, the shortlist, the scorecard, the screener builder, settings, monitor)	PAGES.md
home: the landing page, the cross-tool lists, Manage tools	HOME.md
a stopped tool's read-only history	STOPPED-TOOLS.md

It owns	Form
the listeners	one port per tool and the desk's own
the login check	the desk cookie (BOX.md §4.3)
the live channel	a per-tool push to open pages

5. When it fails

Event	What happens
the front door is down	the desk keeps scanning and photographing; only the pages are out of reach (DESK.md §6)
a page would run heavy work	it is a job or runs in pieces, so one page cannot hold up another tool's scans (DESK.md §4.2)
the desk restarts	the listeners come back on every port; each tool's list comes back from its file; an open page reconnects to the live channel
a request carries no valid login	it is refused by the login check, unless it comes from the box itself

6. Built on

The old desk: every tool served its own pages on its own port; the pages are in legacy/screeners/PAGES.md, the routes in legacy/screeners/API.md, the login and the one address in legacy/architecture/ARCHITECTURE.md.

What failed there:

ID	Fault	Prevented by
D-S15	the Screener tab did not refresh itself; cards were as old as the last load	the live channel pushes changes to open pages
S-1	a tool's port was reachable from the internet (seen in qp's log too)	the login check on every request
—	each tool loaded the whole program to serve its own port	one program, one listener per port, one copy of the code

7. Status

Part	Status
the login and the one address	**built**; the domain and closing the other ports are not done (ARCHITECTURE §9)
one listener per tool port, in one program	**to build** (step B): the way one program holds several tools is built; the tools still run as separate programs
the live channel to open pages	**to build** (step E: live pages)